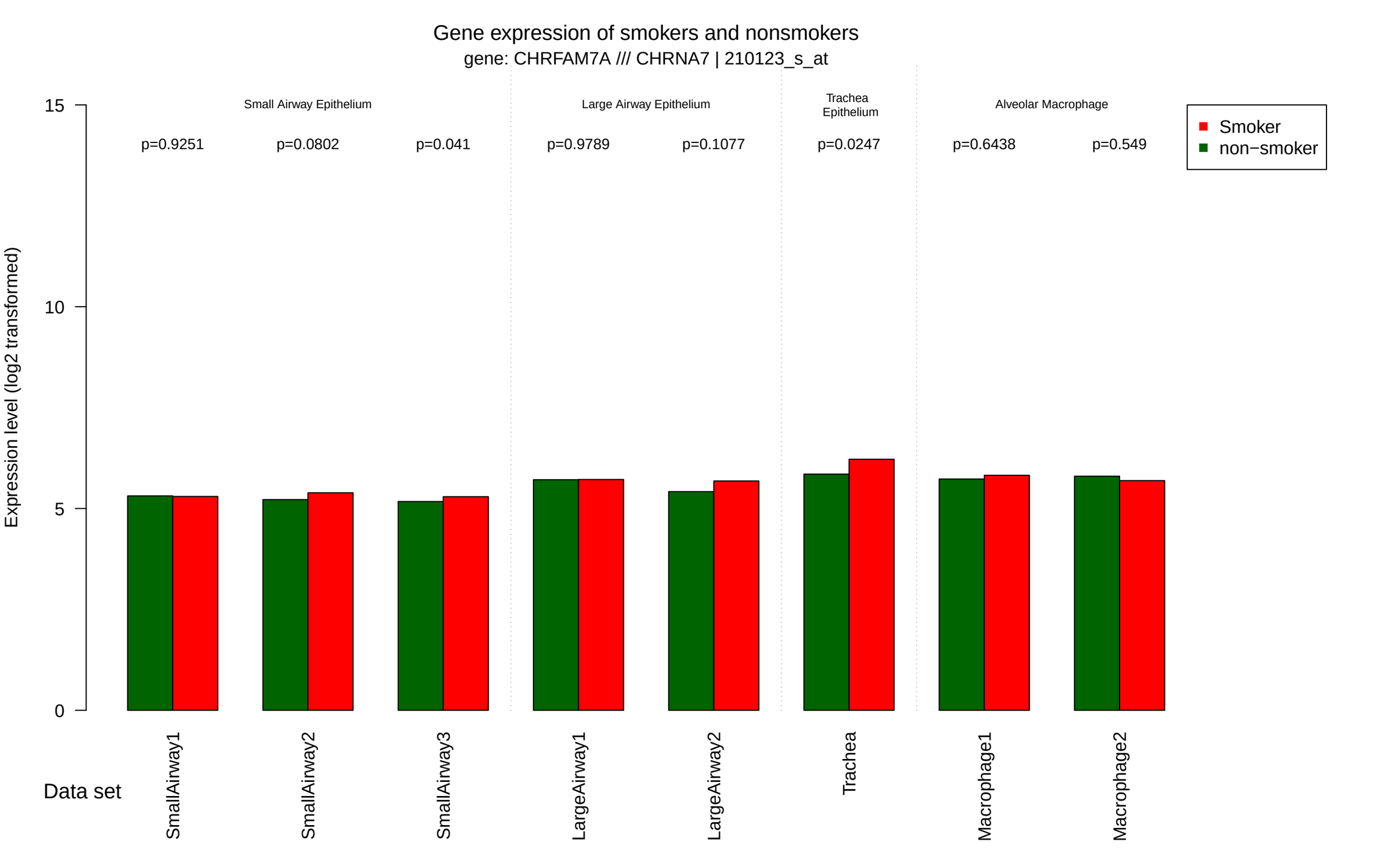
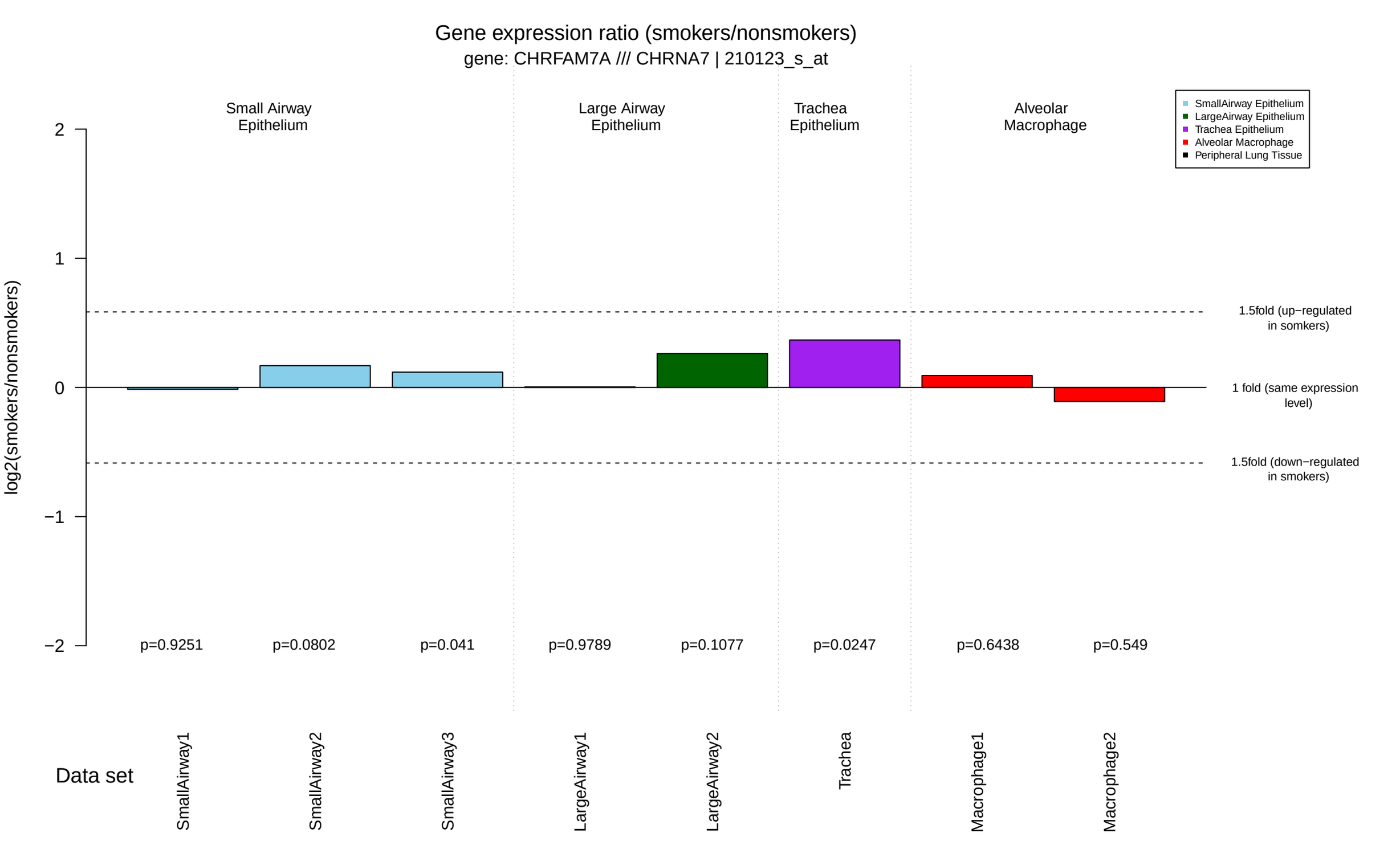
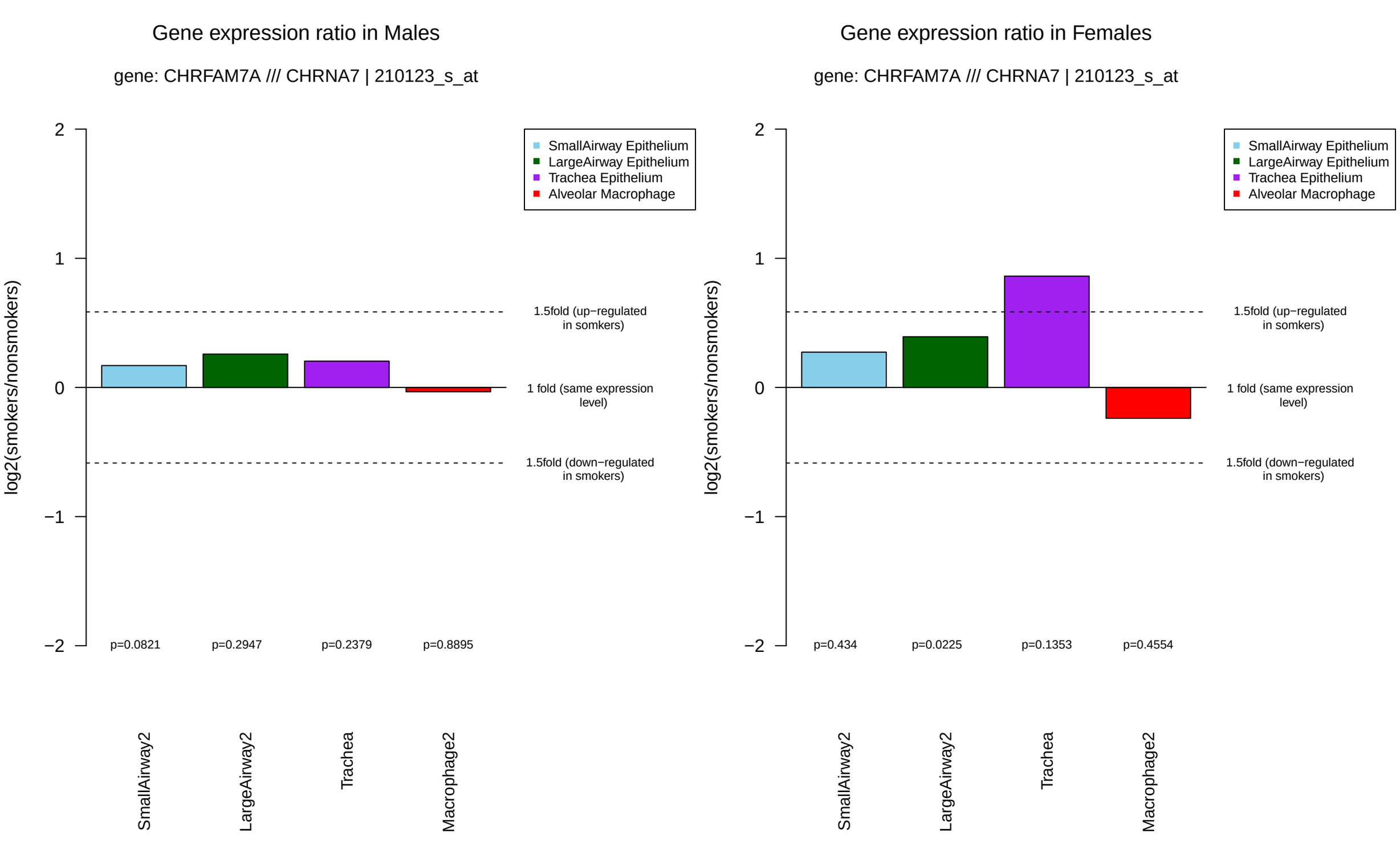
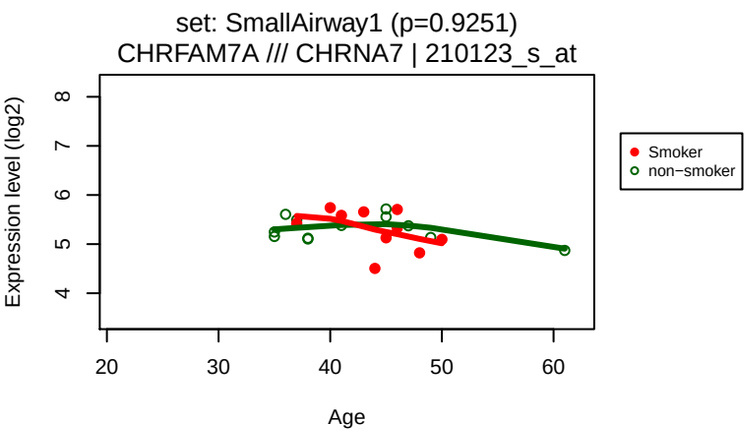
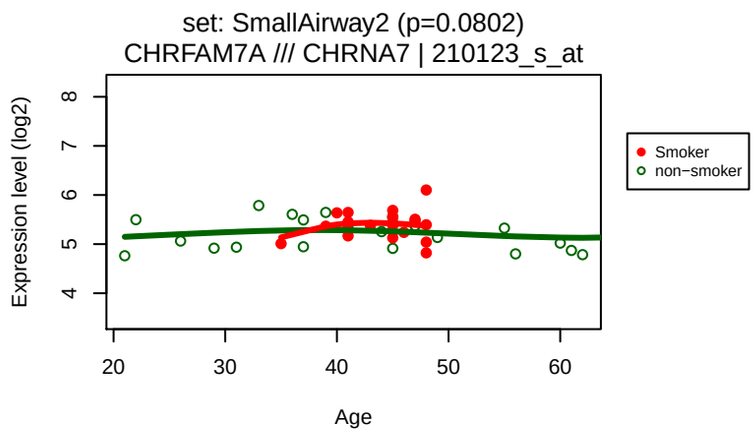
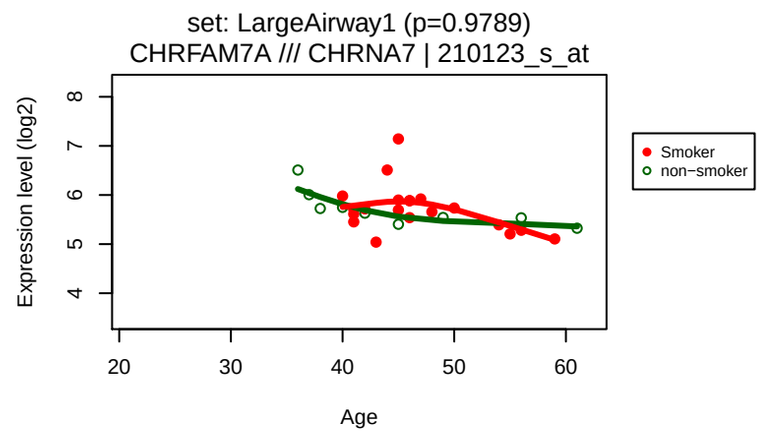

Gene expression of smokers and nonsmokers
gene: CHRFAM7A /// CHRNA7 | 210123_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9251
p=0.0802
p=0.041
p=0.9789
p=0.1077
p=0.0247
p=0.6438
p=0.549
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

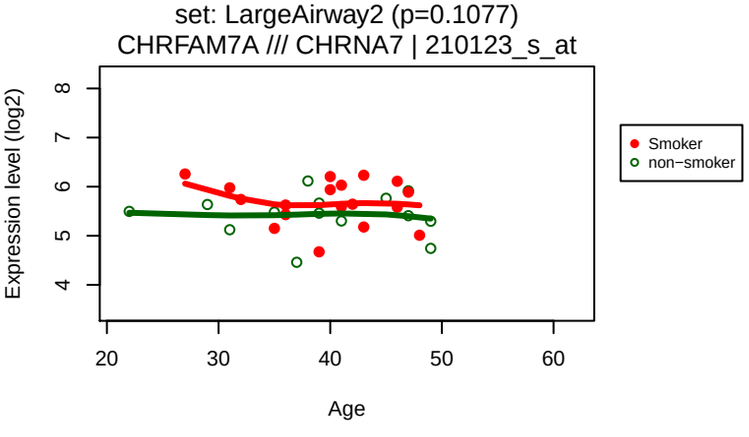

Gene expression ratio (smokers/nonsmokers)
gene: CHRFAM7A /// CHRNA7 | 210123_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9251
p=0.0802
p=0.041
p=0.9789
p=0.1077
p=0.0247
p=0.6438
p=0.549
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

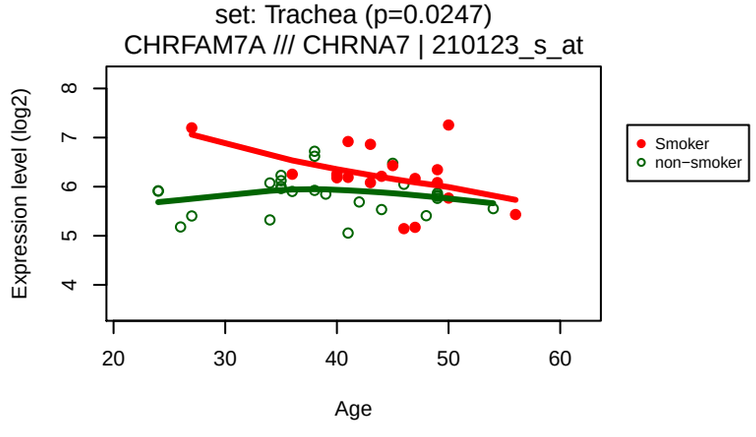

Gene expression ratio in Males
gene: CHRFAM7A /// CHRNA7 | 210123_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0821
p=0.2947
p=0.2379
p=0.8895
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRFAM7A /// CHRNA7 | 210123_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.434
p=0.0225
p=0.1353
p=0.4554
SmallAirway2
LargeAirway2
Trachea
Macrophage2

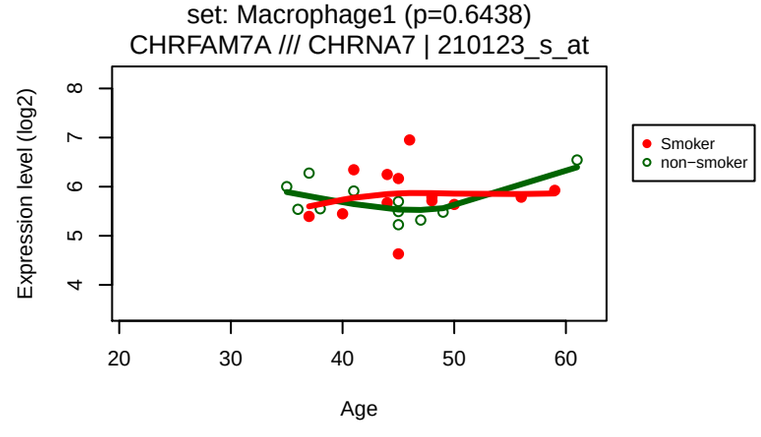

set: SmallAirway1 (p=0.9251)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker

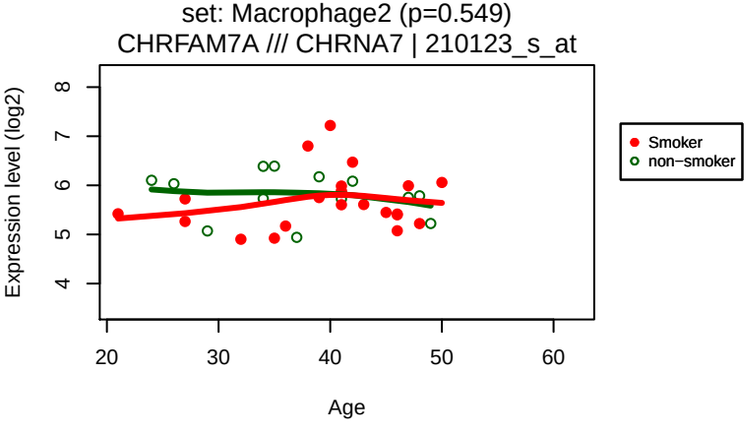

set: SmallAirway2 (p=0.0802)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker

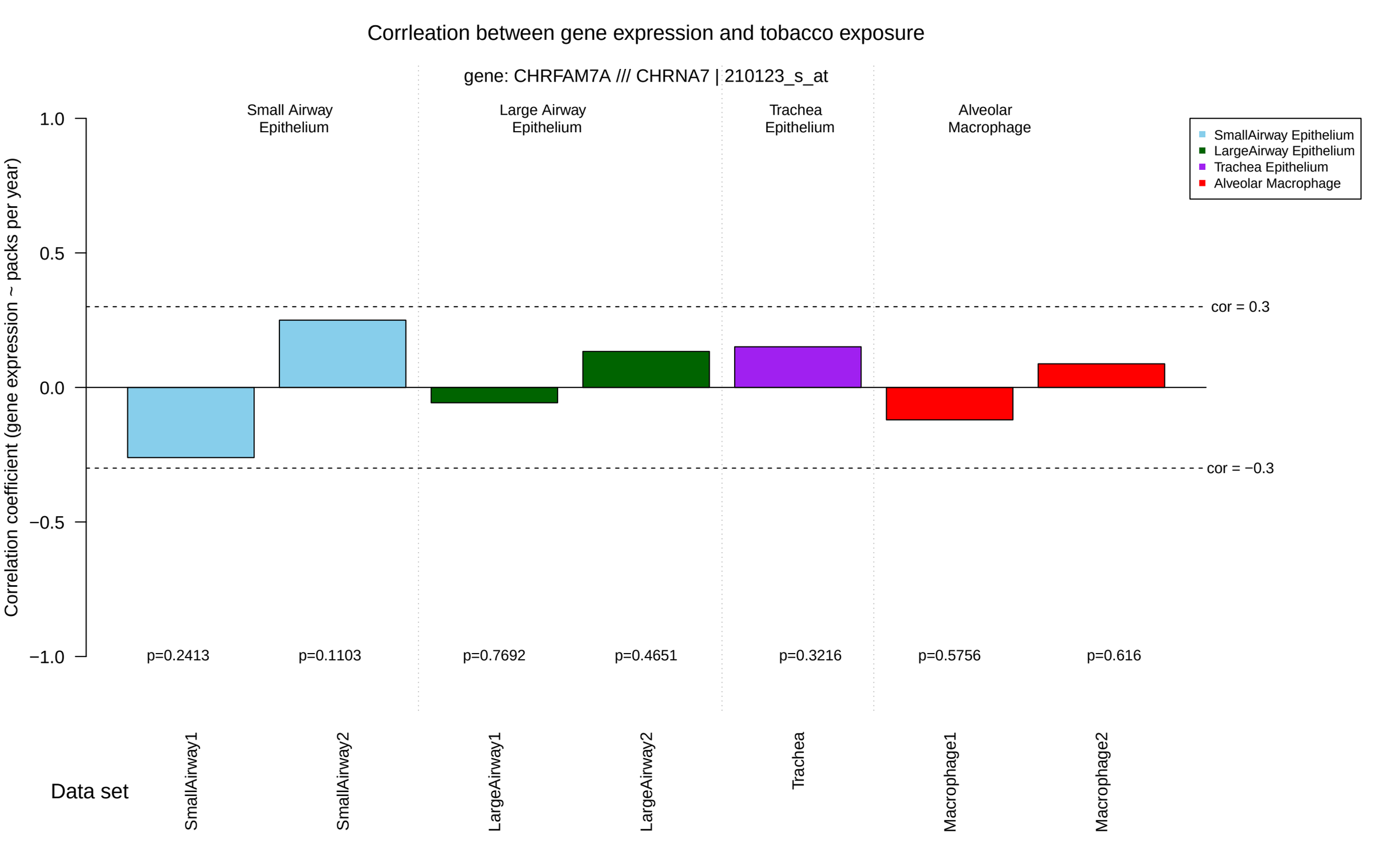

set: LargeAirway1 (p=0.9789)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1077)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0247)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6438)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.549)
CHRFAM7A /// CHRNA7 | 210123_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRFAM7A /// CHRNA7 | 210123_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2413
p=0.1103
p=0.7692
p=0.4651
p=0.3216
p=0.5756
p=0.616
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2